CSE447 | Introduction to Natural Language Processing

HOME ASSIGNMENT (HA) • ALL FIVE MODULES

Course Level	Credit Units	Total Marks	Course Weight
UG	3	10	10%

Student Name: _____ Student ID: _____

Objective: Apply NLP concepts through explanations, calculations, diagrams, practical examples, and a simplified application design.

Question 1 — Module I: Sound | 2 Marks | CLO1

A. Speech Recognition Pipeline (0.5)

Draw: Speech input → Digital audio → Feature extraction → HMM → Viterbi Algorithm → Recognized text. Briefly explain the roles of HMM and Viterbi.

B. Argmax Calculation (0.5)

Given Cat = 0.15, Car = 0.70, Card = 0.15, find the maximum probability and the Argmax result. Explain the difference.

C. HMM / Viterbi Calculation (1)

Path A: previous score 0.80, transition 0.70, emission 0.90. Path B: previous score 0.60, transition 0.50, emission 0.90. Calculate each new score using Previous Score × Transition × Emission. Assuming both paths reach the same hidden state at the same time step, identify which path Viterbi retains and explain why.

Question 2 — Module II: Words and Word Forms | 2 Marks | CLO2

A. Morphological Analysis (0.5)

For Unhappiness, Replaying, Played, and Teacher, identify root, prefix (if any), and suffix (if any). Explain how this helps NLP.

B. Inflection and Derivation (0.5)

Define both terms and give two examples of each. Classify: Play → Played; Teach → Teacher; Book → Books; Happy → Happiness.

C. Finite-State Morphology (1)

Design an FSM recognizing Play, Plays, Played, and Playing. Show states and transitions; label start and accepting states; explain recognition; give one morphological analysis and one generation example.

Question 3 — Module III: Structures | 2 Marks | CLO2 & CLO3

A. CFG and Parsing (1)

For "The student reads a book", use S → NP VP; NP → Det N; VP → V NP; Det → the | a; N → student | book; V → reads. Draw a constituency parse tree, identify subject NP and VP, and explain how top-down parsing begins.

B. Attachment Ambiguity (0.5)

For "I saw the man with a telescope", explain two interpretations and identify what "with a telescope" attaches to in each.

C. Probabilistic Parsing (0.5)

Candidate A = 0.35 and Candidate B = 0.65. Apply Argmax and explain why probabilistic parsing is useful when multiple grammatical interpretations exist.

Question 4 — Module IV: Meaning | 2 Marks | CLO3 & CLO4

A. Word Sense Disambiguation (0.5)

Explain "bank" in (1) "She deposited money at the bank" and (2) "They sat beside the river bank". How does context help?

B. Semantic Roles (0.5)

For "Sara opened the door with a key", identify Agent, Patient, and Instrument. Explain why semantic roles matter.

C. WordNet and Coreference (1)

Define synset; give one hypernym and one synonym example. For "Maya bought a laptop. She uses it every day", identify the referents of "She" and "it", and explain the value of coreference resolution.

Question 5 — Module V: Web 2.0 Applications | 2 Marks | CLO5

A. Sentiment Analysis (0.5)

For "The screen is excellent, but the battery is terrible", identify the sentiment for each aspect. Explain why a single overall label can lose information.

B. Text Entailment (0.5)

Premise: "Sara bought a red car." Classify as entailment, contradiction, or neutral, with reasons: (1) "Sara bought a car." (2) "Sara bought a blue car." (3) "Sara drives to university every day." Assume the same purchase and car.

C. Multilingual NLP Application (1)

Design a university assistant that answers an Arabic question using English university documents. Draw and explain: Arabic question → CLIR → retrieve English documents → question answering → supported answer in Arabic (multilingual generation or machine translation).

Submission & Academic Requirements

- Submit one PDF containing all five questions; include your name, student ID, course code, and assignment title.
- Use your own words. Show calculations and intermediate steps, and include the ASR, FSM, parse-tree, and multilingual application diagrams.
- Include at least five credible academic references; prioritize recent publications (2024–2026). Use IEEE numbered in-text citations and a reference list.
- Use Zotero, Mendeley, EndNote, or another reference manager to organize and format references. Verify bibliographic details and DOIs.
- AI tools are permitted as support. Include a short AI Usage Declaration identifying tools, prompts, and how outputs were verified.
- Be ready to explain your calculations, diagrams, selected publications, publication venues, and relevance during instructor discussion. Journal quartiles may be checked using SJR or JCR (with source/year).
- Python or other programming tools are optional; if used, include screenshots and brief explanations.
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Marking Rubric

Question	Assessment focus	Marks
Q1 — Sound	ASR, Argmax, HMM/Viterbi	2
Q2 — Words and Word Forms	Morphology, inflection/derivation, FSM	2
Q3 — Structures	CFG parse tree, ambiguity, Argmax	2
Q4 — Meaning	WSD, semantic roles, WordNet, coreference	2
Q5 — Applications	Sentiment, entailment, multilingual design	2
TOTAL		10

Evaluation: Full marks require accurate concepts, calculations, diagrams, and explanations. Partial marks apply to incomplete or partially correct responses.